

RSVP Cosmology

*Field Equations, Admissibility, and the Geometry of
Distinction*

Flyxion

Independent Researcher

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This monograph derives the Relativistic Scalar–Vector Plenum (rsvp) field equations from a variational principle, builds the admissibility-manifold hierarchy $M_{\text{adm}} \rightarrow M_{\text{proj}} \rightarrow M_{\text{obs}}$ over rsvp’s state space, and works cosmological structure formation, cosmic memory, and observational diagnosis as instances of the resulting formalism. It is a technical companion to *The Smoothness Paradox*, which develops the same framework’s narrative and cosmological-history consequences; this volume supplies, and proves, the machinery that narrative assumes.

Contents

Preface	xi
I The Plenum and Its Distinctions	1
1 Distinction Before Field	3
1.1 What a Cosmological Theory Is For	3
1.2 Distinctions, Not Particles	3
1.2.1 The Ontological Commitment	3
1.2.2 Why This Is Not Merely Relabeling	3
1.3 Reading This Book	3
1.3.1 Relation to <i>The Smoothness Paradox</i>	3
1.3.2 Relation to <i>Quadratic Gravity as a Derived Interface Theory</i>	4
1.3.3 Organization of the Remaining Parts	4
2 The Scalar–Vector–Entropy Plenum	5
2.1 Field Content	5
2.1.1 The Scalar Field Φ : Distinction Density	5
2.1.2 The Vector Field v : Admissible Transition	5
2.1.3 The Entropy Field S : Accumulated Loss	6
2.2 Why Three Fields and Not One	6
2.3 The Plenum as Continuous Substrate	7
2.3.1 Against the Particles-in-Spacetime Picture	7
2.3.2 What Survives from General Relativity	7
3 Variational Formulation	9
3.1 The RSVP Action	9
3.1.1 Kinetic and Coupling Terms	9
3.1.2 The Role of the Entropy Field in the Action	10
3.2 Field Equations	10
3.2.1 Derivation via the Euler–Lagrange Equations	10
3.2.2 Conserved Quantities and Noether Currents	11

3.3	Hamiltonian Formulation	12
3.3.1	Phase Space and Constraints	12
3.3.2	Relation to the Admissibility Apparatus of Chapter 4	12
4	Admissible Evolution	13
4.1	The Admissibility Relation on Plenum Configurations	13
4.1.1	Admissibility as a Constraint on the Plenum's Own Dynamics, Not an External Postulate	14
4.2	The Admissibility Manifold Hierarchy	14
4.2.1	M_{adm} : The Admissibility Manifold of the Plenum	14
4.2.2	M_{proj} : Projection and Gauge Redundancy	14
4.2.3	M_{obs} : The Observable Cosmological Sector	14
4.3	Entropy as Loss of Admissible Reachability	15
4.3.1	Restating S in Admissibility Terms	15
4.3.2	Relation to, and Departure from, Thermodynamic Entropy	15
II	Distinction, Memory, and Repair	17
5	TARTAN: Recursive Coarse-Graining	19
5.1	The Coarse-Graining Problem in a Distinction-Primary Theory	19
5.2	The TARTAN Operator	19
5.2.1	Scale Consistency	19
5.2.2	Fixed Points and Recursive Structure	19
5.3	TARTAN and the Turnover Surface	19
5.3.1	Statement of the Connection	19
5.3.2	Observable-Equivalence Quotient	20
5.3.3	What This Book Proves and What It Cites	20
6	Distinguishability and the Persistence Cone	21
6.1	Physical Memory as Preserved Inferential Accessibility	21
6.2	The Persistence Cone	21
6.2.1	Physical Cones Versus Observable Cones	22
6.2.2	True Erasure Versus Reconstruction-Bottleneck Failure	22
6.3	Worked Case: Tidal Torque and Galactic Spin [1]	23
6.3.1	Summary of the Empirical Result	23
6.3.2	The Persistence Cone for Protohalo Angular Momentum	24

6.4	Persistence Cones and the RSVP Entropy Field	25
6.4.1	Relating C_ϵ to S	25
7	Repair-Warrant Diagnostics for Cosmological Reconstruction	27
7.1	Diagnosis Before Inference	28
7.2	The Reference Class for a Cosmological Reconstruction	28
7.2.1	$R_T(h)$: Independence from the Object Under Evaluation	28
7.2.2	Grounding the Reference Class in Theoretically Well-Understood Quantities	28
7.3	Structural Distance and the Repair-Warrant Objective	28
7.3.1	δ_{dep} as a Calibrated Structural Distance	28
7.3.2	$J(a h)$: When Intervention Is Warranted	28
7.4	Worked Example: The Initial Power Spectrum as a Diagnostic Proxy [2]	28
7.4.1	Summary of the Case Study	28
7.4.2	What the Case Study Confirms About the Framework's Limits	29
7.5	Non-Implication Propositions	30
III	Structure, History, and Observation	33
8	Structure Formation as Admissible Evolution	35
8.1	From Field Equations to Structure	35
8.2	Structure as Locally Stable Distinction	35
8.2.1	Galaxies, Clusters, and Voids as Admissibility Basins	35
8.3	Comparison with Λ CDM Structure Formation	35
8.3.1	Where the Predictions Agree	35
8.3.2	Where They Diverge and Why	35
9	The Distinguishability Spectrum and the Turnover Surface	37
9.1	Statement of the Distinguishability Spectrum	37
9.2	Statement of the Turnover Surface	37
9.3	The Three-Gate Reknitting Criterion	37
9.3.1	Dynamical Instability	37
9.3.2	Recursive Accessibility	37
9.3.3	Nonlinear Persistence	37

9.4	Operational Equivalence of Terminal and Primordial Smoothness	37
9.5	What This Chapter Assumes From <i>The Smoothness Paradox</i> . . .	38
10	Quadratic Gravity as the Plenum’s Observable Interface	39
10.1	Ghosts, ordinarily	39
10.2	The Hidden Admissibility Direction in the RSVP Plenum . . .	39
10.3	Observable Equivalence Below and Above the Admissibility Scale	39
10.3.1	The Derived Interface Conjecture, Specialized to RSVP .	39
11	Observational Signatures and Tests	41
11.1	Falsifiable Predictions on Record	41
11.2	Predictions Distinguishing RSVP Cosmology from Λ CDM . .	42
11.3	Diagnostic-Coordinate Signatures in Survey Data	42
11.3.1	What a Stage-IV Survey Pipeline Would Need to Report	42
11.4	Open Observational Programs	43
11.4.1	Testing the Persistence Cone in Other Observables . . .	43
11.4.2	Distinguishing Admissible-Level from Observable-Level Defects	43
IV	Open Problems	45
12	What Remains Conjectural	47
12.1	Unproved Claims Carried Forward From This Book	47
12.2	The Relation Between TARTAN Fixed Points and Reknitting .	47
12.3	Toward a Complete Proof of the Derived Interface Conjecture	47
12.4	Directions for Further Work	47
12.5	Conclusion	47
Mathematical Preliminaries		49
12.6	Differential Geometry Conventions	49
12.7	Variational Calculus on the Plenum	49
Notation and Conventions		51
12.8	Field and Admissibility Notation	51
12.9	Relation to Notation in <i>The Smoothness Paradox</i> and <i>Quadratic Gravity as a Derived Interface Theory</i>	51

Selected Derivations	53
12.10 Full Derivation of the RSVP Field Equations	53
12.11 Derivation of the Admissibility Manifold Projections	53
Bibliography	55

List of Figures

Preface

This is a technical book. *The Smoothness Paradox* argues, at the level of cosmic history, that the universe runs smooth to differentiated to smooth, and asks what it would mean for primordial and terminal smoothness to be operationally equivalent rather than a mere repetition of appearances. That argument rests on machinery it largely takes as given: a plenum of scalar, vector, and entropy fields; a notion of admissible evolution; a way of measuring how much of a past configuration remains distinguishable in a present one; and a recursive coarse-graining operation, TARTAN, that relates descriptions of the same history at different scales. This book derives that machinery from first principles, states it as precisely as the subject allows, and proves what can currently be proved about it.

The organizing commitment is the same one that runs through the rest of this research program: distinction is prior to object, admissibility is prior to prediction, and repair is prior to correctness. Applied to cosmology, this means treating the universe not as a fixed background populated by evolving matter fields, but as a plenum whose only physical content is the pattern of distinctions it currently supports, together with the constraints that govern which future patterns of distinction remain reachable from it. What is usually called a particle, a galaxy, or an epoch of cosmic history is, in this reading, a locally stable and locally reconstructible distinction within that plenum — something that persists, within limits, because the dynamics keep re-deriving it from its neighborhood rather than because it is a separately existing thing carried along by the dynamics.

Three bodies of prior work are drawn on directly and cited rather than re-derived where they already exist in adequate form: *Quadratic Gravity as a Derived Interface Theory*, for the admissibility-manifold hierarchy and the argument that observable gravitational physics is a projection from a larger admissibility manifold rather than a complete description of it; the diagnostic-coordinates and repair-warrant apparatus developed across the reconstruction/repair cluster, for the formal treatment of when a departure from a reference class licenses intervention; and *Primordial Distinctions and Cosmo-*

logical Memory, for the persistence-cone construction and its empirical anchor in galactic spin alignment. *The Smoothness Paradox*'s distinguishability spectrum, turnover surface, and three-gate reknottting criterion are stated and used in Part III of this book but not re-proved here; the reader wanting their full narrative and historical treatment should consult that monograph directly.

Flyxion

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Part I

The Plenum and Its Distinctions

Chapter 1

Distinction Before Field

1.1 What a Cosmological Theory Is For

1.2 Distinctions, Not Particles

1.2.1 The Ontological Commitment

1.2.2 Why This Is Not Merely Relabeling

1.3 Reading This Book

1.3.1 Relation to *The Smoothness Paradox*

The Smoothness Paradox is the narrative treatment. It follows cosmic history from primordial regularity through structure formation and runaway smoothing to the question of whether terminal and primordial smoothness are operationally equivalent, and it does so in prose with supporting derivations. This book is the formal counterpart. It derives the rsvp field equations from a variational principle, builds the admissibility hierarchy $M_{\text{adm}} \rightarrow M_{\text{proj}} \rightarrow M_{\text{obs}}$, and develops the coarse-graining operator TARTAN and the persistence cone (Part II), which the narrative uses as vocabulary. It does not retell the smooth-to-differentiated-to-smooth arc.

The division of labor is deliberate and has one exception to note. The distinguishability spectrum, the turnover surface, and the three-gate reknitting criterion are stated and used in Part III of this book but are not re-proved here. Their full development belongs to the narrative volume, and Chapter 6 and the closing chapters list what they assume from it. In the other direction, the memory chapters of the narrative volume rely on the persistence cone defined here. Both volumes hold to the same reading of what a surviving

correlation licenses. The persistence cone is defined through mutual information $I(D; O_{r,t})$, so membership in the cone is a statement about statistical distinguishability of an ensemble. It is not a statement that an individual primordial configuration can be recovered from a present observation, and the worked case of galactic spin in Chapter 6 is presented in that light.

1.3.2 Relation to *Quadratic Gravity as a Derived Interface Theory*

1.3.3 Organization of the Remaining Parts

Chapter 2

The Scalar–Vector–Entropy Plenum

2.1 Field Content

The RSVP plenum is specified by three fields over a base manifold. Unlike the particles-in-spacetime picture, where matter fields are distinct from the geometric background, the plenum is the fundamental object: its field content encodes the pattern of distinctions currently supported and the constraints on which future patterns remain reachable.

2.1.1 The Scalar Field Φ : Distinction Density

The scalar field $\Phi : M \rightarrow \mathbb{R}$ assigns to each point of the plenum manifold M a measure of local distinction density. Where Φ varies rapidly, the plenum supports fine-grained distinctions; where Φ is nearly constant, the local structure is smooth.

Definition 2.1 (Distinction density). Let $U \subset M$ be a region of the plenum. The distinction density in U is

$$\rho_{\text{dist}}(U) = \int_U |\nabla \Phi|^2 dV,$$

the integrated squared gradient of the scalar field over U .

High ρ_{dist} corresponds to a region where many locally stable configurations (galaxies, structures, gradients) coexist; low ρ_{dist} corresponds to regions approaching local equilibrium.

2.1.2 The Vector Field v : Admissible Transition

The vector field $v : M \rightarrow TM$ specifies the direction and rate of admissible evolution at each point. Unlike the Hamiltonian vector field of classical

mechanics, v is constrained by the admissibility relation: only transitions $x \rightarrow y$ satisfying the field equations and boundary conditions of Chapter 3 contribute to v .

Definition 2.2 (Admissible flow). A curve $\gamma : [0, T] \rightarrow M$ is an admissible flow if

$$\frac{d\gamma}{dt} = v(\gamma(t))$$

and $\gamma(t_2) \in \mathcal{A}(\gamma(t_1))$ for all $0 \leq t_1 < t_2 \leq T$, where $\mathcal{A}(x)$ denotes the set of configurations admissible from x .

Not every smooth curve is an admissible flow; v encodes the constraint that evolution must preserve the plenum’s internal consistency conditions.

2.1.3 The Entropy Field S : Accumulated Loss

The entropy field $S : M \rightarrow \mathbb{R}_{\geq 0}$ tracks the accumulated loss of recoverable distinction along an admissible flow. Unlike thermodynamic entropy (a measure of microstate multiplicity), S measures how much of a past configuration’s structure is no longer inferentially accessible from the present one.

Definition 2.3 (Entropy as inferential loss). Let $\gamma : [0, t] \rightarrow M$ be an admissible flow. The entropy accumulated along γ is

$$S(\gamma(t)) - S(\gamma(0)) = \int_0^t (1 - I(\gamma(0); \gamma(s) \mid \gamma(t))) ds,$$

where $I(\gamma(0); \gamma(s) \mid \gamma(t))$ is the mutual information between the initial configuration $\gamma(0)$ and intermediate state $\gamma(s)$, conditioned on the current state $\gamma(t)$.

When $I(\gamma(0); \gamma(s) \mid \gamma(t)) \rightarrow 0$ for some $s < t$, the intermediate configuration $\gamma(s)$ is inferentially inaccessible from $\gamma(t)$: the structure it supported is no longer recoverable. The entropy field integrates this loss.

2.2 Why Three Fields and Not One

General relativity uses a single field (the metric $g_{\mu\nu}$) to encode both geometry and dynamics. Why does RSVP require three?

The answer is that distinction, evolution, and loss are conceptually independent even when dynamically coupled. A configuration can support high distinction density ($|\nabla\Phi|$ large) while evolving slowly ($|\mathbf{v}|$ small) and accumulating entropy slowly (dS/dt small) — or any other combination. Encoding all three in a single field would artificially couple degrees of freedom that the physics keeps separate.

Proposition 2.4 (Independence of field sectors). *There exist admissible flows $\gamma_1, \gamma_2 : [0, T] \rightarrow M$ such that:*

1. $\rho_{\text{dist}}(\gamma_1(t)) = \rho_{\text{dist}}(\gamma_2(t))$ for all t , but $S(\gamma_1(T)) \neq S(\gamma_2(T))$;
2. $S(\gamma_1(t)) = S(\gamma_2(t))$ for all t , but $|\mathbf{v}(\gamma_1(T))| \neq |\mathbf{v}(\gamma_2(T))|$.

Proof sketch. For (1), consider two flows with identical distinction density profiles but different coarse-graining histories: γ_1 passes through a highly differentiated intermediate state that is subsequently smoothed, while γ_2 maintains constant distinction density. Both have the same Φ profile, but γ_1 has accumulated more entropy because its intermediate distinctions are no longer recoverable.

For (2), consider flows in which entropy accumulation saturates (all recoverable distinctions have been lost, $dS/dt = 0$) but the admissible-flow vector $\mathbf{v}$ remains nonzero: the plenum continues to evolve even though no new information is being lost. $\square$ $\square$

2.3 The Plenum as Continuous Substrate

2.3.1 Against the Particles-in-Spacetime Picture

2.3.2 What Survives from General Relativity

Chapter 3

Variational Formulation

3.1 The RSVP Action

The dynamics of the RSVP plenum are derived from a variational principle. We construct an action functional $\mathcal{S}[\Phi, \mathbf{v}, S]$ whose critical points yield the field equations.

3.1.1 Kinetic and Coupling Terms

The RSVP action is

$$\begin{aligned}\mathcal{S}[\Phi, \mathbf{v}, S] &= \int_{\mathcal{M}} \mathcal{L}(\Phi, \mathbf{v}, S, \nabla\Phi, \nabla\mathbf{v}, \nabla S) d^4x, \\ \mathcal{L} &= \mathcal{L}_\Phi + \mathcal{L}_{\mathbf{v}} + \mathcal{L}_S + \mathcal{L}_{\text{coupling}},\end{aligned}\tag{3.1}$$

where $\mathcal{M}$ is the spacetime manifold and the Lagrangian density decomposes into sector-specific terms plus coupling.

Scalar sector:

$$\mathcal{L}_\Phi = -\frac{1}{2} \nabla^\mu \Phi \nabla_\mu \Phi - V(\Phi),$$

where $V(\Phi)$ is a self-interaction potential. We take $V(\Phi) = \frac{\lambda}{4} \Phi^4 + \frac{m^2}{2} \Phi^2$ for concreteness, though the framework allows other forms.

Vector sector:

$$\mathcal{L}_{\mathbf{v}} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu},$$

where $F_{\mu\nu} = \nabla_\mu v_\nu - \nabla_\nu v_\mu$ is the field strength of the vector field $\mathbf{v}$.

Entropy sector:

$$\mathcal{L}_S = -\frac{1}{2} \nabla^\mu S \nabla_\mu S.$$

The entropy field has no self-interaction potential; it is driven entirely by coupling to Φ and $\mathbf{v}$.

3.1.2 The Role of the Entropy Field in the Action

The coupling term is

$$\mathcal{L}_{\text{coupling}} = -\alpha S \nabla^\mu \Phi \nabla_\mu \Phi - \beta S F^{\mu\nu} F_{\mu\nu},$$

where $\alpha, \beta > 0$ are coupling constants. This form ensures:

1. S increases when Φ or v exhibit large gradients or curvature;
2. The entropy sector acts as a sink for distinguishability: high S suppresses further differentiation.

3.2 Field Equations

3.2.1 Derivation via the Euler–Lagrange Equations

The field equations are obtained by demanding $\delta S = 0$ under variations $\delta\Phi$, δv^μ , δS that vanish on the boundary.

Scalar field equation:

$$\square\Phi - V'(\Phi) = 2\alpha\nabla^\mu(S\nabla_\mu\Phi), \quad (3.2)$$

where $\square = \nabla^\mu \nabla_\mu$ is the d'Alembertian and $V'(\Phi) = dV/d\Phi$. The right-hand side is the back-reaction from the entropy field: high S damps scalar-field gradients.

Vector field equation:

$$\nabla_\mu F^{\mu\nu} = 2\beta\nabla_\mu(SF^{\mu\nu}). \quad (3.3)$$

This is a modified Maxwell equation with an entropy-dependent source term.

Entropy field equation:

$$\square S = \alpha\nabla^\mu\Phi\nabla_\mu\Phi + \beta F^{\mu\nu}F_{\mu\nu}. \quad (3.4)$$

The entropy field is sourced by gradients in Φ and curvature in v , with no self-damping: once distinction is lost, S does not spontaneously decrease.

3.2.2 Conserved Quantities and Noether Currents

The action (3.1) is invariant under spacetime translations $x^\mu \rightarrow x^\mu + a^\mu$. By Noether's theorem, this implies a conserved stress-energy tensor.

Theorem 3.1 (Energy-momentum conservation). *The stress-energy tensor*

$$T^{\mu\nu} = \nabla^\mu \Phi \nabla^\nu \Phi + F^{\mu\lambda} F^\nu_\lambda + \nabla^\mu S \nabla^\nu S - g^{\mu\nu} \mathcal{L} \quad (3.5)$$

satisfies $\nabla_\mu T^{\mu\nu} = 0$ on solutions to the field equations (3.2)–(3.4).

Proof. Standard Noether procedure: compute $\delta\mathcal{S}$ under $x^\mu \rightarrow x^\mu + \epsilon a^\mu$, use the field equations to eliminate $\delta\Phi$, δv^μ , δS , and integrate by parts. The boundary term vanishes by assumption, leaving $\nabla_\mu T^{\mu\nu} = 0$. $\square$ $\square$

Note 3.2 (Open: Distinction-count or admissibility-measure invariants). Energy-momentum conservation is a standard geometric result. Whether the RSVP action admits a further conserved current — specifically, a dynamical invariant measuring distinction count, admissibility measure, or inferential capacity that is preserved by the field equations themselves — remains to be derived.

Such an invariant would have to emerge from a symmetry of the action beyond spacetime translations. Candidates include:

- A global $U(1)$ phase symmetry of Φ (if the scalar has complex structure not yet specified);
- A coarse-graining symmetry relating TARTAN scales;
- An invariant constructed from the admissibility relation itself, not merely from the field content.

The title of this research program, "Numerical Claims Are Conserved Quantities," suggests such an invariant exists, but its explicit form and derivation are not yet on record. This is flagged for completion in Part IV.

3.3 Hamiltonian Formulation

3.3.1 Phase Space and Constraints

3.3.2 Relation to the Admissibility Apparatus of Chapter 4

Chapter 4

Admissible Evolution

4.1 The Admissibility Relation on Plenum Configurations

Definition 4.1 (Parameterized admissible transition). Let Γ be a constraint structure specifying:

1. The RSVP field equations (Chapter 3);
2. Boundary conditions (initial data, asymptotic conditions);
3. Conservation laws (energy-momentum, gauge constraints).

A configuration y is admissible from x relative to Γ , written $x \rightsquigarrow_{\mathcal{A}} y$ (with Γ clear from context), if there exists a solution of the field equations connecting x to y that satisfies all constraints in Γ .

Remark 4.2 (Admissibility is relative to constraints). Admissibility is always admissibility *relative to a constraint structure*. Different constraint structures yield different admissible configuration spaces:

$$\Gamma_1 \neq \Gamma_2 \implies M_{\text{adm}}(\Gamma_1) \neq M_{\text{adm}}(\Gamma_2)$$

in general. When we write $x \rightsquigarrow_{\mathcal{A}} y$, the constraint structure Γ should be either explicit or clear from context. In this monograph, unless stated otherwise, Γ refers to the standard RSVP constraints: field equations (3.2)–(3.4), local energy-momentum conservation, and causality.

4.1.1 Admissibility as a Constraint on the Plenum's Own Dynamics, Not an External Postulate

4.2 The Admissibility Manifold Hierarchy

4.2.1 M_{adm} : The Admissibility Manifold of the Plenum

Definition 4.3 (Admissibility manifold). Given a constraint structure Γ , the admissibility manifold is

$$M_{\text{adm}}(\Gamma) = \{x \in \mathcal{X} : x \text{ satisfies all constraints in } \Gamma\},$$

where $\mathcal{X}$ is the full configuration space of $(\Phi, \mathbf{v}, S)$ fields over the base manifold.

The admissibility manifold is the space of all configurations consistent with the plenum's own dynamics. Not every element of $\mathcal{X}$ lies in $M_{\text{adm}}(\Gamma)$: arbitrary field configurations need not solve the field equations or respect boundary conditions.

4.2.2 M_{proj} : Projection and Gauge Redundancy

Definition 4.4 (Projection manifold). Let $\sim_{\text{gauge}}$ be the equivalence relation identifying gauge-equivalent configurations. The projection manifold is the quotient

$$M_{\text{proj}} = M_{\text{adm}}(\Gamma) / \sim_{\text{gauge}} .$$

Gauge redundancy includes coordinate transformations, overall phase rotations of Φ , and other symmetries of the action that do not affect physical predictions. M_{proj} contains only physically distinguishable configurations.

4.2.3 M_{obs} : The Observable Cosmological Sector

Definition 4.5 (Observable manifold). Let $\pi : M_{\text{proj}} \rightarrow M_{\text{obs}}$ be the measurement map that assigns to each gauge-equivalence class of configurations the set of its observable predictions (galaxy counts, CMB multipoles, correlation functions, etc.). The observable manifold is

$$M_{\text{obs}} = \pi(M_{\text{proj}}).$$

Critically, π is neither injective nor surjective in general:

- Not injective: distinct admissible configurations can yield identical observable predictions (underdetermination);
- Not surjective: not every element of M_{obs} corresponds to an admissible configuration (over-parameterization).

The failure of injectivity is the source of the diagnostic gap explored in Chapter 7: observable agreement does not imply admissible-level correctness.

4.3 Entropy as Loss of Admissible Reachability

4.3.1 Restating S in Admissibility Terms

4.3.2 Relation to, and Departure from, Thermodynamic Entropy

Part II

Distinction, Memory, and Repair

Chapter 5

TARTAN: Recursive Coarse-Graining

5.1 The Coarse-Graining Problem in a Distinction-Primary Theory

5.2 The TARTAN Operator

Definition 5.1 (TARTAN coarse-graining). $\mathcal{T}_{\text{TARTAN}}$ maps a plenum configuration and a scale parameter to a coarser configuration, subject to the requirement that admissibility relations at the coarser scale are induced by, and consistent with, admissibility relations at the finer scale.

5.2.1 Scale Consistency

5.2.2 Fixed Points and Recursive Structure

5.3 TARTAN and the Turnover Surface

5.3.1 Statement of the Connection

The turnover surface Σ_{turn} is the locus in (r, t) space where the distinguishability spectrum $\mathcal{D}(r, t)$ reaches its maximum and begins to decrease. TARTAN coarse-graining induces a natural map from fine-scale to coarse-scale admissibility relations; the turnover surface marks where this map begins to merge previously-distinct configurations.

5.3.2 Observable-Equivalence Quotient

Definition 5.2 (Observable-equivalence relation). Two configurations $x, y \in \mathcal{X}$ are observably equivalent if

$$x \sim_o y \iff \forall O \in \mathcal{O}_{\text{adm}} : O(x) = O(y),$$

where $\mathcal{O}_{\text{adm}}$ is the set of admissible observables.

The equivalence class $[x]_{\sim_o}$ is the fiber of the projection map $\pi : M_{\text{proj}} \rightarrow M_{\text{obs}}$. This equivalence is preserved under further projections — but only tautologically: it is preserved because it is defined as the fiber.

Remark 5.3. The observable-equivalence quotient $\mathcal{X}/\sim_o$ is not a dynamical invariant in the sense of a Noether current. It is preserved by construction (quotients are well-defined), not by the field equations. Its role is to ensure that TARTAN’s coarser-scale admissibility relations are induced by, rather than merely consistent with, finer-scale ones.

5.3.3 What This Book Proves and What It Cites

This chapter establishes:

- The TARTAN operator’s scale-consistency requirement (Definition 5.1);
- The observable-equivalence quotient construction (Definition 5.2);
- That observable equivalence is preserved under projection (Remark 5.1).

This chapter cites but does not re-derive:

- The connection between TARTAN fixed points and the reknotting criterion (open problem, see Chapter 12);
- The turnover surface’s detailed structure (see *The Smoothness Paradox*).

Chapter 6

Distinguishability and the Persistence Cone

6.1 Physical Memory as Preserved Inferential Accessibility

6.2 The Persistence Cone

Definition 6.1 (Persistence cone). Given a primordial distinction D , an observable O , a reconstruction procedure $\mathcal{R}$, and a threshold $\epsilon > 0$, the persistence cone is

$$C_\epsilon(D) = \{(O, r, t) : I(D; O_{r,t}) > \epsilon\},$$

the region of observable/scale/time space in which D remains statistically distinguishable above threshold, where $I(D; O_{r,t})$ is the mutual information between the primordial distinction D and the observable O measured at co-moving scale r and cosmic time t .

The persistence cone is not a light cone. Distinctions can be causally accessible (lie within the past light cone) yet inferentially inaccessible (lie outside the persistence cone) if the intervening dynamics have smoothed them below the reconstruction threshold.

Proposition 6.2 (Persistence cone containment). *For fixed distinction D and observable O :*

1. *The persistence cone is bounded: $C_\epsilon(D) \subseteq [r_{\min}, r_{\max}] \times [t_0, t_{\text{eras}}]$ for some $r_{\min}, r_{\max}, t_{\text{eras}} < \infty$.*
2. *If $\epsilon_1 < \epsilon_2$, then $C_{\epsilon_2}(D) \subseteq C_{\epsilon_1}(D)$ (higher thresholds yield smaller cones).*

3. The persistence cone shrinks under coarse-graining: if $\mathcal{R}_{\text{coarse}}$ averages over a scale Δr , then $C_\epsilon(D; \mathcal{R}_{\text{coarse}}) \subseteq C_\epsilon(D; \mathcal{R}_{\text{fine}})$ for $\mathcal{R}_{\text{fine}}$ resolving scales $\ll \Delta r$.

Proof sketch. (1) Boundedness follows from the second law: $I(D; O_{r,t})$ is non-increasing in t for fixed r (information is lost, not spontaneously created), so the cone must pinch off at some t_{eras} where $I < \epsilon$ for all scales. The scale bounds $r_{\text{min}}, r_{\text{max}}$ follow from finite correlation lengths.

(2) Immediate from the definition: $\{I > \epsilon_2\} \subseteq \{I > \epsilon_1\}$ when $\epsilon_2 > \epsilon_1$.

(3) Coarse-graining reduces mutual information by the data-processing inequality: $I(D; \mathcal{R}_{\text{coarse}}(O)) \leq I(D; O)$. $\square$

6.2.1 Physical Cones Versus Observable Cones

The persistence cone defined above is an *observable* cone: it measures what can be reconstructed from a specific observable O using a specific procedure $\mathcal{R}$. The *physical* persistence cone, denoted $C_{\epsilon\text{phys}}(D)$, is the union over all possible observables and optimal reconstruction:

$$C_{\epsilon\text{phys}}(D) = \bigcup_{O, \mathcal{R}} C_\epsilon(D; O, \mathcal{R}).$$

In practice, $C_{\epsilon\text{phys}}$ is inaccessible: we never have access to all observables or optimal reconstruction. What we measure is always some projection $C_{\epsilon\text{obs}} \subseteq C_{\epsilon\text{phys}}$.

6.2.2 True Erasure Versus Reconstruction-Bottleneck Failure

A distinction D is *truly erased* at time t if $(O, r, t) \notin C_{\epsilon\text{phys}}(D)$ for all observables O and scales r : no procedure, no matter how clever, can recover D from the state at time t .

A distinction experiences *reconstruction-bottleneck failure* at t if $(O, r, t) \notin C_{\epsilon\text{obs}}(D)$ for the observable and procedure actually used, but $(O', r', t) \in C_{\epsilon\text{phys}}(D)$ for some other (O', r') : the distinction is still physically present, but the chosen observable cannot see it.

Example 6.3 (Galactic spin as a persistence-cone observable). Sheng et al. [1] compare projected spin directions of present-day central galaxies, measured from integral-field velocity maps (MaNGA), with directions predicted from a primordial tidal field reconstructed by the ELUCID constrained simulation. Their strongest sample, gas in 781 central elliptical galaxies, shows a

weighted mean directional correlation of 0.13 at a smoothing scale of $3.4 h^{-1} \text{ Mpc}$. The significance against a null that shuffles observed spins within halo-mass bins is 7.0σ when the smoothing scale is optimized for the data only, 6.3σ at a fixed scale, and 5.9σ when each null realization also optimizes its scale.

In persistence-cone language: the primordial distinction D (the orientation structure of the tidal field acting on a protohalo, as reconstructed from the initial conditions) remains recoverable above threshold in the observable $O = \text{projected galaxy spin direction}$, read out through the reconstruction procedure $\mathcal{R}$ of the ELUCID pipeline. The detection is evidence that this $(O, \mathcal{R})$ lies in the cone for this D . It is evidence at the level of the ensemble. With a correlation of order 0.1, the observed spin of an individual galaxy leaves most of the uncertainty about its own primordial environment in place, so the result establishes persistence of statistical information and not recoverability of individual states.

6.3 Worked Case: Tidal Torque and Galactic Spin [1]

6.3.1 Summary of the Empirical Result

The data are MaNGA integral-field velocity maps for central galaxies inside the volume where the ELUCID constrained simulation supplies initial conditions. Kinematic position angles are fitted separately for ionized gas and stars, giving projected spin directions. The predicted direction for each galaxy is computed from the reconstructed tidal field acting on its protohalo, after Lagrangian-space remapping. The correlation statistic is a weighted mean of the cosine between observed and predicted directions, with weights favoring protohaloes whose tidal twist is strong. The strongest signal is gas in the 781 central ellipticals (0.13 at $3.4 h^{-1} \text{ Mpc}$). Central spirals (815 with reliable gas spins) show much weaker correlations, and stellar spins are generally less well correlated than gas spins in both morphologies.

The significance is reported three ways (7.0σ , 6.3σ , 5.9σ , see the example above), all against a null that permutes observed spins within halo-mass bins. The lowest mass bin, where reconstruction and remapping errors dominate, is excluded from the combined figures, and the authors describe the combined values as approximate because weak covariance between bins may

remain. The paper attributes the elliptical-spiral difference to several contributors acting together: higher host masses, gas position angles measured with smaller uncertainty than stellar ones, and a residual difference at fixed halo mass whose origin it leaves open. It does not show that ellipticals conserve primordial angular momentum more faithfully.

6.3.2 The Persistence Cone for Protohalo Angular Momentum

To place the measurement in the cone $C_\epsilon(D) = \{(O, r, t) : I(D; O_{r,t}) > \epsilon\}$ the four ingredients must be identified. The primordial distinction D is the orientation structure of the tidal field acting on the protohalo, summarized by the eigenframe of the tidal tensor and the associated tidal twist. The observable O is the projected spin direction of the central galaxy, read separately from the gas and the stellar velocity fields, which gives two observables O_{gas} and O_{stars} with different cones. The reconstruction procedure $\mathcal{R}$ is the ELUCID reconstruction of the initial conditions together with Lagrangian-space remapping. The scale coordinate r is the smoothing scale applied to the reconstructed field, whose optimum is reported to scale with the halo's Lagrangian radius above roughly $10^{12.5} M_\odot$, and the time coordinate is the present.

Three statements follow, each at the strength the data support.

1. **Membership.** For gas in central ellipticals, $(O_{\text{gas}}, r_{\text{opt}}, t_0)$ is in the cone at the ensemble level. The measured quantity is a weighted mean directional correlation, not a mutual information, so the threshold ϵ is not fixed by the data. Converting the correlation to $I(D; O)$ requires a model of the joint distribution of predicted and observed directions.
2. **Low-mass haloes.** In the lowest halo-mass bin the paper finds little or no excess correlation and attributes this to reconstruction and remapping errors. In the vocabulary of the previous section this is consistent with reconstruction-bottleneck failure. The data cannot separate it from true erasure, because both produce no detectable correlation through the same pipeline.
3. **Gas versus stars.** A smaller correlation for O_{stars} than for O_{gas} would suggest nested cones, $C_\epsilon(D; O_{\text{stars}}) \subseteq C_\epsilon(D; O_{\text{gas}})$. The paper reports

the ordering qualitatively and notes that gas position angles are fitted with smaller uncertainties, so measurement noise alone could reproduce part of it. Nesting is therefore a hypothesis here, not a result.

What the example cannot supply is the state-level converse. Membership of (O, r, t_0) in $C_\epsilon(D)$ says the ensemble of spins carries information about the ensemble of tidal environments. It does not say that a single galaxy's spin narrows its own primordial environment appreciably, and a correlation of order 0.1 leaves that narrowing weak.

6.4 Persistence Cones and the RSVP Entropy Field

6.4.1 Relating C_ϵ to S

Chapter 7

Repair-Warrant Diagnostics for Cosmological Reconstruction

7.1 Diagnosis Before Inference

7.2 The Reference Class for a Cosmological Reconstruction

7.2.1 $R_{\Gamma}(h)$: Independence from the Object Under Evaluation

7.2.2 Grounding the Reference Class in Theoretically Well-Understood Quantities

7.3 Structural Distance and the Repair-Warrant Objective

7.3.1 δ_{dep} as a Calibrated Structural Distance

7.3.2 $J(a \mid h)$: When Intervention Is Warranted

7.4 Worked Example: The Initial Power Spectrum as a Diagnostic Proxy [2]

7.4.1 Summary of the Case Study

Hoellinger and Leclercq (2024) [2] address forward-model misspecification in Stage-IV galaxy survey inference. Their SELFI/ABC-PMC pipeline recon-

constructs the initial matter power spectrum θ from simulated galaxy count power spectra, then uses that reconstruction as a diagnostic proxy before committing to the expensive cosmological-parameter inference.

The case study instantiates the repair-warrant framework as follows:

- **Object under evaluation:** $h = \gamma$, the SELFI posterior mean for the initial power spectrum, obtained from a possibly-misspecified forward model of the survey.
- **Reference class:** $R_\Gamma(h) = P(\theta)$, a Gaussian prior with mean and covariance obtained by propagating CMB-informed cosmological priors through a Boltzmann solver. This has the required independence: it is built from theory and prior experiments, not from the survey simulator under evaluation.
- **Structural distance:** The Mahalanobis distance

$$d_M(\gamma, \theta_0 \mid S) = \sqrt{(\gamma - \theta_0)^\top S^{-1}(\gamma - \theta_0)},$$

calibrated against the null distribution obtained from 5,000 samples drawn from $P(\theta)$.

- **Diagnostic results:** Under correct model specification, $d_M(\gamma, \theta_0 \mid S)$ lies within the expected range and the inferred cosmological parameters (Ω_m, σ_8) are unbiased. Under misspecification (incorrect galaxy bias, halved gravity-solver time steps, inaccurate light-cone treatment), d_M exceeds the 95th percentile of the null distribution while the cosmological parameters are biased by more than 2σ — yet the raw galaxy count power spectra differ by only $\sim 1\%$, making the bias nearly invisible without the diagnostic stage.

The two-step architecture (reconstruct θ , check it against $P(\theta)$, only then infer ω) is diagnosis before commitment: a cheap proxy sensitive to systematic effects is evaluated before the expensive cosmological inference is run.

7.4.2 What the Case Study Confirms About the Framework's Limits

Two features of the Hoellinger-Leclercq results confirm limitations the abstract framework states explicitly rather than optimistically eliding:

Observable projections are alarms, not verdicts. The paper warns that “a larger distance does not necessarily imply a worse model, but hints at the presence of systematic effects.” This is the Mahalanobis distance functioning as , an observable projection of the underlying admissibility-level departure . A large licenses investigation but does not certify which mechanism is responsible or how large the true structural damage is. The follow-up work (their Section IV.A.4) matches characteristic scales in k -space to distinguish among candidate systematic effects — diagnosis of the second, harder kind: determining where in intervention space the responsible structure sits, not merely that $\neq 0$.

Constraint satisfaction at one level does not imply admissibility at another. Some tested systematic effects (misspecified masked point sources) have negligible effect on the reconstructed θ , while others (halved gravity-solver time steps) produce inverse errors in θ exceeding the forward error in the summary statistics by a wide margin. The galaxy count power spectra pass (forward error $< 1\%$), yet the initial power spectrum reconstruction is rejected at nearly 2σ — a clean empirical instance of the non-implication proposition: passing a check phrased at the wrong level offers no assurance about the level that matters.

7.5 Non-Implication Propositions

The repair-warrant framework establishes several *non*-implications — things that do not follow from diagnostic evidence, even when that evidence is statistically significant.

Proposition 7.1 (Observable agreement does not imply structural correctness).

$$(h, R_{\Gamma}(h)) = 0 \not\Rightarrow (h, R_{\Gamma}(h)) = 0.$$

Proof sketch. The measurement map $\pi : M_{\text{proj}} \rightarrow M_{\text{obs}}$ is a projection, not an isometry. Let $\ker(\pi) = \{x \in M_{\text{proj}} : \pi(x) = 0\}$ be the kernel. For any $h \in M_{\text{proj}}$ and $k \in \ker(\pi)$, we have $\pi(h + k) = \pi(h)$, so $(\pi(h), \pi(h + k)) = 0$, yet $(h, h + k)$ can be arbitrarily large. $\square$

Proposition 7.2 (Diagnostic discrepancy does not imply unique diagnosis).

$$(h, R_{\Gamma}(h)) > \epsilon \not\Rightarrow |\{D : \text{defect } D \text{ produces discrepancy}\}| = 1.$$

Proof sketch. The preimage $\pi^{-1}(\{\mathcal{O}(h)\})$ is generically not a singleton. Multiple distinct defects in this fiber produce identical observable signatures yet differ at the admissible level. $\square$

Proposition 7.3 (Diagnostic discrepancy does not imply unique repair).

$$(h, R_{\Gamma}(h)) > \epsilon \not\Rightarrow \exists! a : J(a | h) < 0.$$

Detection alone does not determine intervention. The repair-warrant objective $J(a | h)$ provides a selection criterion among a given candidate set $\mathcal{A}(h)$, but generating that candidate set with attributable, disentangled effects is the hard unsolved part.

Remark 7.4. Hoellinger and Leclercq’s Section IV.A.4 demonstrates this third non-implication empirically: multiple systematic effects (galaxy bias, light-cone approximation, gravity-solver time steps) produce similar Mahalanobis distances. Diagnosing which defect is responsible requires matching characteristic scales in k -space — a second diagnostic beyond the scalar distance d_M .

Part III

**Structure, History, and
Observation**

Chapter 8

Structure Formation as Admissible Evolution

8.1 From Field Equations to Structure

8.2 Structure as Locally Stable Distinction

8.2.1 Galaxies, Clusters, and Voids as Admissibility Basins

8.3 Comparison with Λ CDM Structure Formation

8.3.1 Where the Predictions Agree

8.3.2 Where They Diverge and Why

Chapter 9

The Distinguishability Spectrum and the Turnover Surface

9.1 Statement of the Distinguishability Spectrum

9.2 Statement of the Turnover Surface

9.3 The Three-Gate Reknottting Criterion

9.3.1 Dynamical Instability

9.3.2 Recursive Accessibility

9.3.3 Nonlinear Persistence

9.4 Operational Equivalence of Terminal and Primal Smoothness

Definition 9.1 (Operational equivalence via interface distance). Two epochs E_1, E_2 are operationally equivalent if

$$d_{\text{int}}(E_1, E_2) = 0,$$

where d_{int} is the interface-native distance (Jensen-Shannon divergence on observable distributions). No admissible measurement distinguishes them.

Remark 9.2. Operational equivalence ($d_{\text{int}} = 0$) is weaker than being the same point in the underlying state space. It means only that the observable projections coincide, not that the admissible-level configurations are identical.

Whether primordial and terminal smoothness also collapse to one point under a TARTAN-induced quotient $\mathcal{X}/\sim_o$ is a separate, stronger claim.

The central question of *The Smoothness Paradox* can now be stated precisely: if the universe evolves from primordial smoothness through differentiation back to terminal smoothness, and if $d_{\text{int}}(E_{\text{primordial}}, E_{\text{terminal}}) = 0$, does this operational equivalence license treating them as the same epoch for purposes of cosmological continuation? Or does the intervening history distinguish them at a level the interface-native distance cannot detect?

9.5 What This Chapter Assumes From *The Smoothness Paradox*

This chapter cites but does not re-derive:

- The distinguishability spectrum $\mathcal{D}(r, t)$ and its relation to TARTAN coarse-graining;
- The turnover surface Σ_{turn} in (r, t) space where TARTAN's recursive description stops differentiating and begins reconverging;
- The three-gate reknotting criterion (dynamical instability, recursive accessibility, nonlinear persistence) and its two onset routes.

Readers requiring the full derivation and narrative treatment should consult *The Smoothness Paradox* directly. This monograph uses those results as established machinery.

Chapter 10

Quadratic Gravity as the Plenum's Observable Interface

10.1 Ghosts, ordinarily

10.2 The Hidden Admissibility Direction in the RSVP Plenum

10.3 Observable Equivalence Below and Above the Admissibility Scale

10.3.1 The Derived Interface Conjecture, Specialized to RSVP

Chapter 11

Observational Signatures and Tests

11.1 Falsifiable Predictions on Record

Currently, one concrete falsifiable prediction exists:

[Persistence-cone gas-stellar asymmetry] The gas spin correlation with the primordial tidal field exceeds the stellar spin correlation:

$$\Pi_{\text{gas}} > \Pi_{\text{stars}}.$$

Empirical status: Sheng et al. [1] find, for both central ellipticals and central spirals, that stellar spins are generally less well correlated with the reconstructed tidal field than gas spins. This is a qualitative pattern in their correlation curves. The paper does not report a separate significance for the gas-stellar difference. The $5.9\text{--}7.0\sigma$ figures quoted there measure the gas correlation in central ellipticals against a shuffled null, not the asymmetry. The paper also notes that gas position angles are fitted with systematically smaller uncertainties than stellar ones, which by itself could produce part of the pattern. The prediction is therefore consistent with the data but not yet tested against the failure condition below.

Failure condition: If the asymmetry disappears after controlling for halo mass and for the difference in position-angle measurement uncertainty between gas and stellar tracers, the dynamical-coherence interpretation of the persistence cone loses support. The framework does not require this specific prediction to hold universally, but its failure in this regime would indicate that gas dynamics do not preserve primordial angular momentum correlations as persistently as the framework currently assumes.

11.2 Predictions Distinguishing RSVP Cosmology from Λ CDM

Note 11.1. Further distinguishing predictions between RSVP and Λ CDM are under development. The framework is not yet rich enough in observational consequences to claim multiple independent tests. Sections below outline directions for future work rather than established predictions.

11.3 Diagnostic-Coordinate Signatures in Survey Data

11.3.1 What a Stage-IV Survey Pipeline Would Need to Report

For the repair-warrant diagnostic apparatus of Chapter 7 to function as an operational check on cosmological inference pipelines, Stage-IV surveys (DESI, Euclid, LSST) would need to report:

1. The reconstructed initial power spectrum θ alongside the inferred cosmological parameters ω ;
2. The Mahalanobis distance $d_M(\gamma, \theta_0 \mid S)$ calibrated against the null distribution from the prior;
3. Characteristic scale information (which k -bins dominate the discrepancy, if any) to enable defect attribution;
4. The candidate set of systematic effects $\mathcal{A}(h)$ that could produce an observed discrepancy, with repair costs estimated.

No current survey pipeline implements this two-step diagnostic architecture as standard practice. Hoellinger and Leclercq (2024) demonstrate its feasibility in a research context, not an operational one.

11.4 Open Observational Programs

11.4.1 Testing the Persistence Cone in Other Observables

The persistence-cone construction (Chapter 6) is not limited to galactic spin. Candidate observables include:

- Filament orientations relative to large-scale tidal shear;
- Void shapes and their alignment with primordial overdensity gradients;
- Higher-order correlation functions sensitive to non-Gaussian initial conditions.

Each provides an independent test of whether primordial distinctions persist in current structure.

11.4.2 Distinguishing Admissible-Level from Observable-Level Defects

Testing whether a forward-model defect biases only observables or propagates to the admissible level requires independent reconstructions of the same quantity from different observables. If galaxy clustering and weak lensing both reconstruct θ with similar discrepancies from $P(\theta)$, the defect is likely admissible-level; if discrepancies differ, the defect may be observable-specific.

Part IV

Open Problems

Chapter 12

What Remains Conjectural

- 12.1 Unproved Claims Carried Forward From This Book**
- 12.2 The Relation Between TARTAN Fixed Points and Reknotting**
- 12.3 Toward a Complete Proof of the Derived Interface Conjecture**
- 12.4 Directions for Further Work**
- 12.5 Conclusion**

Mathematical Preliminaries

12.6 Differential Geometry Conventions

12.7 Variational Calculus on the Plenum

Notation and Conventions

12.8 Field and Admissibility Notation

12.9 Relation to Notation in *The Smoothness Paradox* and *Quadratic Gravity as a Derived Interface Theory*

Selected Derivations

12.10 Full Derivation of the RSVP Field Equations

12.11 Derivation of the Admissibility Manifold Projections

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